Module 3 Homework 2 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 89,991 of 99,990 points. 90%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1 Jenny has taken 5 tests for her geometry class and has an average score of 83. If she wants an average score of 85, what does she need to get on her next test?

✓	**Your Answer**	95
	Correct Answer	95

Explanation

What can we find out about Jenny's past scores, given her average score? That's right, we can figure out the total number of points she earned. She took 5 tests and her average was a 83, so she must have had $5 \times 83 = 415$ points in total. She is taking one more test, and she wants an 85 average, so she wants a total of $6 \times 85 = 510$ points. The difference, $510 - 415 = 95$, is the score Jenny needs on the next test.

2 If number a decreases by 20% and number b increases by 10%, then which of the following expressions is equal to the sum of a and b after they have changed?

Your Answer	$0.8a + 1.1b$
Correct Answer	$0.8a + 1.1b$

Explanation

We take 20% of a away from a, which leaves us with

$$a - 0.20a = (1 - 0.20)a = 0.80a$$

and we add 10% of b to the number b, which means b turns into

$$b + 0.10b = 1.1b$$

The sum of these is $0.8a + 1.1b$, answer choice number four.

3 The average of Andrew, Beatrice, and Carl's ages is 29. What will the average of their ages be in exactly 4 years?

✓	**Your Answer**	33
	Correct Answer	33

Explanation

For the sake of generality, you could suppose that Andrew, Beatrice, and Carl are all 29 years old. It doesn't affect the average if we arbitrarily set their ages in this way. Then, in 4 years, each of them will be $29 + 4 = 33$ years old, which is the same as their average, since they are all the same age.

4 If an arithmetic sequence has a as its first term, which of the following is the nth term of this sequence? (Here r is the difference between neighboring terms.)

Your Answer	nth term $= a + (n-1)r$
Correct Answer	nth term $= a + (n-1)r$

Explanation

The first term of the arithmetic sequence is a, and the difference between neighboring terms is r, so the next few terms of the sequence look like this:

$$\begin{aligned} 2^{\text{nd}} &: a + r \\ 3^{\text{rd}} &: a + r + r \\ 4^{\text{th}} &: a + r + r + r \\ 5^{\text{th}} &: a + r + r + r + r \end{aligned}$$

If we combine the r's, we see that

$$\begin{aligned} 2^{\text{nd}} &: a + 1r \\ 3^{\text{rd}} &: a + 2r \\ 4^{\text{th}} &: a + 3r \\ 5^{\text{th}} &: a + 4r \end{aligned}$$

If we think about how the numbers in an arithmetic sequence grow, they grow based on the value of n, so as n grows bigger and bigger, if you double n, then the value of the term will approximately double as well. This is not exactly true because if the value of a is large compared to r, then for small r, it isn't true that the sequence grows proportional to n. This is the reason why the choices that contain a^n or r^n are incorrect, because those choices grow much faster with n. You can also think of arithmetic sequences as growing similar to the points on a line.

5 Find the value of $\sqrt{(\sqrt{7})^3}$ to the power 4.

Your Answer	[no answer given]
Correct Answer	343

Explanation

The important thing to remember is that when you take an exponent and put it to another power, the two powers multiply together. One way to do this is to think of $(\sqrt{7})^3$ as the square root of $(\sqrt{7})^6$. Then

$$\begin{aligned}\left(\sqrt{(\sqrt{7})^3}\right)^4 &= \left((\sqrt{7})^3\right)^2 \\ &= \sqrt{7}^6 \\ &= 7^3 \\ &= 343\end{aligned}$$

Another way to do this is to use fractional exponents.

$$\begin{aligned}\left(((\sqrt{7})^3)^{\frac{1}{2}}\right)^4 &= (\sqrt{7}^3)^2 \\ &= 343\end{aligned}$$

6 The simplified version of $\sqrt{12+6\sqrt{3}}$ should have which of the following forms, where a and b are both integers, and the square root is in simplest form?

Your Answer	$a+b\sqrt{3}$
Correct Answer	$a+b\sqrt{3}$

Explanation

We saw that in the video lesson, if we let x to equal the expression $\sqrt{12+6\sqrt{3}}$, then we can square both sides to get $12+6\sqrt{3}$. Then, we can guess that x might have an integer term plus a term with a square root of 3, in other words that

$$x = a + b\sqrt{3},$$

because then if you square x, the cross terms will give a $\sqrt{3}$ term, and the other terms will be simply integers:

$$\begin{aligned} x^2 &= (a+b\sqrt{3})^2 \\ &= (a+b\sqrt{3})(a+b\sqrt{3}) \\ &= a^2 + ab\sqrt{3} + ab\sqrt{3} + (\sqrt{3})^2 \\ &= a^2 + 3 + 2ab\sqrt{3} \end{aligned}$$

7 Find the fraction that is the same as $0.133133133\ldots$ Your answer should be a fraction, without spaces, e.g. 1/2

✓	**Your Answer**	133/999
	Correct Answer	133/999

Explanation

Repeating decimals are nasty, so let's try to write equations that don't have any repeating decimals at all! Here x can be shifted by multiplying by 1000 to get $133.133133133\ldots$, which looks a lot like x. If we write an equation, we can express the relationship between this new number and x.

$$133.133133133\ldots = 1000x$$

We just need to find another way to express x, and if we look at $133.133133133\ldots$, this will hopefully stand out to us!

$$133.133133133\ldots = 133 + x$$

Now, we have two equations, which means that we don't need the troublesome number, the repeating decimal $133.133133\ldots$.

$$\begin{aligned} 1000x &= 133 + x \\ 999x &= 133 \\ x &= \frac{133}{999} \end{aligned}$$

8 When a decimal point is placed between two consecutive numbers, the new number that is formed is their average. What is the sum of the two numbers?

✓	**Your Answer**	9
	Correct Answer	9

Explanation

Since we are averaging two whole numbers, we know that their average has to have $.5$ at the end of it. Since the digits of the average are the digits of the two numbers, then the second number must be 5. This means that the other number must be either a 4 or 6, since the two numbers are consecutive. We can simply check the average that you get from the two possibilities: $\frac{6+5}{2} = 5.5$ or $\frac{4+5}{2} = 4.5$. The first case doesn't work, so the two numbers must be 4 and 5. We want their sum, so that is $4 + 5 = 9$.

9 Simplify this expression: $\sqrt{12+6\sqrt{3}}$

Your Answer	$3+\sqrt{3}$
Correct Answer	$3+\sqrt{3}$

Explanation

From the video lesson, we saw that we can guess that the answer is in the form of an integer plus a term with a square root of 3. Then, we can write $a+b\sqrt{3}$ for the value that we want to figure out.

$$\begin{aligned} a+b\sqrt{3} &= \sqrt{12+6\sqrt{3}} \\ (a+b\sqrt{3})^2 &= \sqrt{12+6\sqrt{3}} \end{aligned}$$

Looking at this, we can see that the $6\sqrt{3}$ term is found from the cross terms that we get from expanding the left side, in other words, $2ab\sqrt{3}$, so we can guess that the values of a and b might be 3 and 1.

$$\begin{aligned} \sqrt{12+6\sqrt{3}} &= \sqrt{9+2\times 3\times\sqrt{3}+(\sqrt{3})^2} \\ &= \sqrt{(3+\sqrt{3})^2} \\ &= 3+\sqrt{3} \end{aligned}$$

10 Given two numbers a and b, where $a \neq b$, what is true about the geometric mean compared to the arithmetic mean of a and b?

Your Answer	The geometric mean is always smaller than the arithmetic mean
Correct Answer	The geometric mean is always smaller than the arithmetic mean

Explanation

The geometric mean of a and b is equal to $\sqrt{a \times b}$. The arithmetic mean of a and b is the simple average, or $\frac{a+b}{2}$. Suppose that the geometric and arithmetic means were equal. Then

$$\begin{aligned} \frac{a+b}{2} &= \sqrt{a \times b} \\ \left(\frac{a+b}{2}\right)^2 &= a \times b \\ (a+b)^2 &= 4ab \\ a^2 + 2ab + b^2 &= 4ab \\ a^2 - 2ab + b^2 &= 0 \\ (a-b)^2 &= 0 \\ &\implies a = b \end{aligned}$$

However, this can't be true, because the problem says that $a \neq b$. There is no other way for $(a-b)^2$ to equal 0, so it's not possible for the geometric mean to equal the arithmetic mean. We can say that the left side of the above equation must be positive, since $(a-b)^2$, being a square, must be positive. The left side of the equation is always larger, which means we can write

$$\frac{a+b}{2} > \sqrt{a \times b}$$

Then the geometric mean is always smaller than the arithmetic mean, so the correct answer is the second choice.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 The third and twenty-seventh terms of an arithmetic sequence are -51 and 21, respectively. What is the seventy-second term?

Your Answer	[no answer given]
Correct Answer	156

Explanation

Let's try to find the common difference between the terms in this sequence. We start from the 3rd term and go to the 27th term, so that means the common difference is added $27 - 3 = 24$ times. Now, let's see how the term itself changes, from -51 to 21, which is a change of $21 - (-51) = 72$. Divide this total change over the number of times to get $\frac{72}{24} = 3$, which is how much you add to a term to get the next term. The 72nd term would be $72 - 27 = 45$ steps away from the 27th term, which is 21, so we should add $45 \times 3 = 135$ to 21, which gives us 156.

12 If one arithmetic sequence has first term a and difference between terms r, and a second arithmetic sequence has first term $a+1$ and difference between terms equal to $r+1$, then find the difference between the sum of the first 2019 terms of the first sequence and the first 2019 terms of the second sequence.

Your Answer	[no answer given]
Correct Answer	$2019+1+2+3+4+\ldots+2017+2018$

Explanation

The sum of the first 2019 terms of the first sequence is

$$2019a+\left(1+2+3+4+\ldots+2017+2018\right)r$$

and to get the sum of the first n terms of the second sequence, we need to replace a with $a+1$ and replace r by $r+1$.

$$2019(a+1)+\left(1+2+3+4+\ldots+2018\right)(r+1)$$

We want to know how much bigger this is compared to the quantity before it, which is

$$2019+\left(1+2+3+4+\ldots+2018\right)$$

Our answer is choice number 3.

13 If $(\frac{1}{4})^4 = 2^x$, find the value of x.

Your Answer	[no answer given]
Correct Answer	-8

Explanation

Since $\frac{1}{4}$ equals 2^{-2}, we can write

$$\begin{aligned}\frac{1}{4}^4 &= (2^{-2})^4 \\ &= 2^{-2\times 4} \\ &= 2^{-8}\end{aligned}$$

14 Which is bigger, $\left(\left((2^2)^2\right)^2\right)^2$ or $2^{2^{2^{2^2}}}$?

Your Answer	[no answer given]
Correct Answer	$2^{2^{2^{2^2}}}$

Explanation

If you use your intuition, you might guess that the second power tower is bigger, because the size is accumulating in the power of 2. In the first power tower, the sizes are growing in the base, which is then put to the 2nd power.

The first should be calculated from the inside out:

$$= \left(\left((2^2)^2\right)^2\right)^2$$

$$= \left(\left((4)^2\right)^2\right)^2$$

$$= \left((16)^2\right)^2$$

$$= 256^2$$

$$= 65536$$

The second should be calculated from the top of the power to the base number:

$$2^{2^{2^{2^2}}}$$

$$= 2^{2^{2^4}}$$

$$= 2^{2^{16}}$$

$$= 2^{65536}$$

15 Find the fractional form of the decimal $0.2\overline{777}$.
Your answer should be a fraction in simplest terms without spaces, e.g. 1/2

Your Answer	[no answer given]
Correct Answer	5/18

Explanation

If you look at this for a very long time, you might notice that $0.2\overline{777}$ is similar to $0.1\overline{666}$. It is

$$0.\overline{111} + 0.1\overline{666}$$

and the decimal $0.1\overline{666}$ is equal to $\frac{1}{6}$. Also, $0.\overline{111}$ is like

$$\begin{aligned}\frac{0.\overline{333}}{3} &= \frac{1}{3} \times \frac{1}{3} \\ &= \frac{1}{9}\end{aligned}$$

This means that we have

$$\frac{1}{9} + \frac{1}{6} = \frac{2}{18} + \frac{3}{18} = \frac{5}{18}.$$

Check that this doesn't simplify, and this is our answer.

16 Find the value of

$$1.142857142857\ldots+2.285714285714\ldots+3.428571428571\ldots+4.571428571428\ldots+5.714285714285\ldots+6.857142857142\ldots+$$
$$7.142857142857\ldots+8.285714285714\ldots+9.428571428571\ldots+10.571428571428\ldots+11.714285714285\ldots+12.857142857142\ldots+$$
$$13.142857142857\ldots+14.285714285714\ldots+15.428571428571\ldots+16.571428571428\ldots+17.714285714285\ldots+18.857142857142\ldots+$$
$$19.142857142857\ldots+20.285714285714\ldots+21.428571428571\ldots+22.571428571428\ldots+23.714285714285\ldots+24.857142857142\ldots$$

Your Answer	[no answer given]
Correct Answer	312

Explanation

Did you notice that there is a pattern in the repeating part of these decimals? We've seen this before! Yes, these are the decimal expansions of fractions of the form $\frac{n}{7}$ again!

$$\begin{aligned}
&.142857142857\ldots+.285714285714\ldots+.428571428571\ldots+.571428571428\ldots+.714285714285\ldots+.857142857142\ldots \\
&= \frac{1}{7}+\frac{2}{7}+\frac{3}{7}+\frac{4}{7}+\frac{5}{7}+\frac{6}{7} \\
&= \frac{1+2+3+4+5+6}{7} \\
&= \frac{21}{7} \\
&= 3.
\end{aligned}$$

This means our sequence is really just

$$\begin{aligned}
&1+2+3+4+5+\ldots+22+23+24+4\times 3 \\
&= \frac{24\times 25}{2}+4\times 3 \\
&= 12\times 25+12
\end{aligned}$$

Whenever we have a 25 in our multiplication, it's nice to go hunting for 4 to times with it. Aha, we have one!

$$\begin{aligned}
&3\times 4\times 25+12 \\
&= 3\times 100+12 \\
&= 312.
\end{aligned}$$

17 Find the value of $(\sqrt[3]{2}+\sqrt[3]{16}+\sqrt[3]{54})^3$.

Your Answer	[no answer given]
Correct Answer	432

Explanation

Let's use the fact that $16 = 2 \times 8$ and $54 = 2 \times 27$. Since 8 and 27 are both cubes, we can take them out of the cube root, leaving a cube root of 2 in every term.

$$\begin{aligned}
\left(\sqrt[3]{2}+\sqrt[3]{16}+\sqrt[3]{54}\right)^3 &= \left(\sqrt[3]{2}+\sqrt[3]{2\times 8}+\sqrt[3]{2\times 27}\right)^3 \\
&= \left(\sqrt[3]{2}+2\sqrt[3]{2}+3\sqrt[3]{2}\right)^3 \\
&= \left((1+2+3)\sqrt[3]{2}\right)^3 \\
&= \left(6\sqrt[3]{2}\right)^3 \\
&= 6^3 \times 2 \\
&= 216 \times 2 \\
&= 432.
\end{aligned}$$

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Simplify this expression: $\sqrt{\sqrt{5}-\sqrt{3-\sqrt{29-12\sqrt{5}}}}$

Your Answer	[no answer given]
Correct Answer	1

Explanation

Start from the innermost square root, $\sqrt{29-12\sqrt{5}}$, and try to find the simplified form of it, $a+b\sqrt{5}$.

$$
\begin{aligned}
&\sqrt{\sqrt{5}-\sqrt{3-\sqrt{29-12\sqrt{5}}}}\\
&=\sqrt{\sqrt{5}-\sqrt{3-\sqrt{(2\sqrt{5})^2-2\times 2\sqrt{5}\times 3+3^2}}}\\
&=\sqrt{\sqrt{5}-\sqrt{3-\sqrt{(2\sqrt{5}-3)^2}}}\\
&=\sqrt{\sqrt{5}-\sqrt{3-(2\sqrt{5}-3)}}\\
&=\sqrt{\sqrt{5}-\sqrt{3-2\sqrt{5}+3}}\\
&=\sqrt{\sqrt{5}-\sqrt{6-2\sqrt{5}\times 1}}\\
&=\sqrt{\sqrt{5}-\sqrt{(\sqrt{5})^2-2\sqrt{5}\times 1+1}}\\
&=\sqrt{\sqrt{5}-\sqrt{(\sqrt{5}-1)^2}}\\
&=\sqrt{\sqrt{5}-(\sqrt{5}-1)}\\
&=\sqrt{\sqrt{5}-\sqrt{5}+1}\\
&=\sqrt{1}\\
&=1
\end{aligned}
$$

19 The sum of the first 4 terms in an arithmetic series is 58. The sum of the first 8 terms in the arithmetic series is 236. What is the first term of the series? Your answer should be a fraction without spaces, e.g. 1/2

✕	**Your Answer**	[no answer given]
	Correct Answer	13/4

Explanation

Let's call the first term f and the common difference between two terms d. Then, the sum of the first 4 terms is

$$\begin{aligned} f + (f + d) + (f + 2d) + (f + 3d) &= 58 \\ 4f + 6d &= 58 \end{aligned}$$

We could write the sum of the first 8 terms by adding up all 8 terms, or we could use the fact that we already have the sum of the first 4 terms:

$$\begin{aligned} \text{sum of first 8 terms} &= 236 \\ \text{sum of first 4 terms} + (f + 4d) + (f + 5d) + (f + 6d) + (f + 7d) &= 236 \\ 4f + 6d + 4f + 4d + 5d + 6d + 7d &= 236 \\ 8f + 28d &= 236 \end{aligned}$$

Let's double the first equation so that we get two equations with $8f$, which will allow us to cancel out the $8f$.

$$4f + 6d = 58$$

and

$$8f + 12d = 116.$$

Subtracting the equations, we get

$$\begin{aligned} 16d &= 120 \\ &= \frac{120}{16} \\ &= \frac{15}{2} \end{aligned}$$

In order to get the value of the first term, f, we can use $4f + 45 = 58$, which will give us $4f = 13$, so the first term is $\frac{13}{4}$.

20 For each of the integers from 1 to 2019, take the greatest positive integer less than or equal to the square root of it. Find the sum of all these numbers. Write the number without commas, e.g. 11999

Your Answer	[no answer given]
Correct Answer	59510

Explanation

Before you run away from this question, try sitting down and writing down a few of the terms, looking for a pattern. Up until 4, which is the square of 2, the square roots of our numbers will round down to 1. From 4 up until 9, which is the square of 3, the square roots of our numbers will round down to 2. And so on and so forth:

$$\begin{aligned} 1,2,3 &\implies 1 \\ 4,5,6,7,8 &\implies 2 \\ 9,10,11,12,13,14,15 &\implies 3 \\ 16,17,18,19,20,21,22,23,24 &\implies 4 \\ &\dots \\ &\dots \\ 1936,1937,1938,\dots,2015,2016,2017,2018,2019 &\implies 44 \\ n^2,n^2+1,n^2+2,n^2+3,\dots,(n+1)^2-1 &\implies n \end{aligned}$$

Do you see a neat pattern here? There are three numbers in the first row, five numbers in the second row, seven numbers in the third row, etc.

We get the numbers in the second-to-last row because $1936 = 44^2$, so it is the first number whose square root rounds down to 44. The next square is 2025, so all of the rest of the numbers up until our last number, 2019, will have their square root round down to 44. However, we don't get to include all the numbers from 44^2 up until right before 45^2, since we end at 2019 instead of 2025, so the last line will not match the pattern that you see in the earlier rows.

Already, we can write a pattern for the numbers up until right before 1936. We have **3 ones, 5 twos, 7 threes, 9 fours, ...** up until some number of 43's. But how do we find how many 43's? (Remember, we are not counting the 44 s right now, since the last row is incomplete and doesn't match the pattern.)

Our pattern looks like this:

$$\Big(3\times 1\Big)+\Big(5\times 2\Big)+\Big(7\times 3\Big)+\Big(9\times 4\Big)+\dots+\Big(?\times 43\Big)$$

$$\Big(1\times 2+1\Big)\times 1+\Big(2\times 2+1\Big)\times 2+\Big(3\times 2+1\Big)\times 3+\dots(43\times 2+1)\times 43$$

where each term is $(2n+1)\times n$ starting from $n=1$ and going until $n=43$. This is the same as the sum of a $2n^2$ sequence added to a n sequence:

$$\begin{aligned} &2\times 1^2+2\times 2^2+2\times 3^2+\dots 2\times 43^2+1+2+3+4+\dots 43 \\ &2\Big(1^2+2^2+3^2+4^2+5^2+6^2+\dots+43^2\Big)+\Big(1+2+3+4+\dots+43\Big) \end{aligned}$$

To find the sum of the first 43 squares, we are going to do something a bit strange that doesn't seem connected to what we're doing, but it gives a relation for the sum of the first 43 squares. Let's suppose that we expand the following polynomial, and try to get the cubic terms on the same side:

$$\begin{aligned} (x-1)^3 &= \\ &= x^3-3x^2+3x-1 \\ x^3-(x-1)^3 &= 3x^2-3x+1 \end{aligned}$$

Suppose we start with $x=1$ and make a new equation for every x counting up from 1 until 43. If we add up all these equations, we get a telescoping sum on the left side!

$$\begin{aligned} &\Big(1^3-(1-1)^3\Big)+\Big(2^3-(2-1)^3\Big)+\dots\Big(43^3-(43-1)^3\Big) \\ &=\Big(1^3-0+2^3-1^3+3^3-2^3+4^3-3^3+\dots 43^3\Big) \\ &=43^3 \end{aligned}$$

On the right side, we have the sum of the right-hand sides of the equation:

$$= 3\left(1^2+2^2+3^2+\ldots 43^2\right) - 3\left(1+2+3+\ldots 43\right) + \left(1+1+\ldots 1\right)$$

Then, let's isolate the sum of the squares, since that is what we are trying to find:

$$\begin{aligned}
3\left(1^2+2^2+3^2+\ldots+43^2\right) &= 43^3 + 3\left(1+2+3+\ldots 43\right) - \left(1+1+1+\ldots+1\right) \\
3\left(1^2+2^2+3^2+\ldots+43^2\right) &= 43^3 + 3\left(1+2+3+\ldots 43\right) - 43 \\
\left(1^2+2^2+3^2+\ldots+43^2\right) &= \frac{1}{3}\left(43^3 + 3\frac{43(43+1)}{2} - 43\right) \\
&= \frac{1}{3}43^3 + \frac{1}{2}43^2 + \frac{43}{2} - \frac{1}{3}43 \\
&= \frac{1}{3}43^3 + \frac{1}{2}43^2 + \frac{1}{6}43 \\
&= 24,434
\end{aligned}$$

In the last line, we used the fact that the sum of the numbers from 1 to 43 is $\frac{43\times 44}{2}$.

Let's remember what we are trying to solve for:

$$\begin{aligned}
2\left(1^2+2^2+3^2+\ldots 43^2\right) + \left(1+2+3+\ldots 43\right) = \\
&= 2\times 24,434 + \frac{43\times 44}{2} \\
&= 54,868 + 946 \\
&= 55,814
\end{aligned}$$

In the last step above, we applied the formula for the sum of the numbers from 1 to 43 again.

Wait a second! We mustn't forget the numbers 1936 through 2019, because those each have their square roots round down to 44. There are $2019 - 1936 + 1 = 84$ of these numbers, so they all contribute $84 \times 44 = 3696$.

Our grand total is $55814 + 3696 = 59510$.

Math note. The sum of the first n squares is usually written in the form

$$\frac{n(n+1)(2n+1)}{6}$$

Summary

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